

Positive Square Energy of Hexacyclic Graphs

Conditional manuscript skeleton—draft, not a theorem announcement

AI-generated draft; no human authorship is claimed

10 August 2026

Draft status. The headline implication in Theorem 1.1 is explicitly conditional. This manuscript does not claim that every hexacyclic graph has the stated property. Promotion is blocked until the order-nine and order-ten exact owners, the rank-six orders-2–10 master, and the all-connected synthesis master all pass their full normal and optimized replay gates.

Abstract

Let G be a finite simple connected graph with n vertices and $n + 5$ edges. This draft records the verified analytic reduction of the desired inequality $s^+(G) \geq n$ to two proof interfaces. The multiblock interface is covered by an existing owner-exact ledger. In the unique positive-rank cyclic-block case, suppression gives one of 1198 rank-six multigraph kernels, with order distribution 1, 4, 26, 84, 216, 314, 325, 162, 66 on two through ten branch vertices. Exact path elimination reduces arbitrary subdivisions to physical parity orbits and canonical-plus-coordinate frontiers, and one-vertex additivity includes arbitrary rooted trees. The final single-block premise is not yet available: in particular, incomplete order-nine or order-ten coverage is not a theorem owner, and the orders-2–10 and all-connected masters are not green. Reproducibility appendices below therefore contain a mixture of verified commands and clearly marked promotion placeholders, not a publication record or theorem certificate.

1 Scope and conditional statement

All graphs in the spectral statements are finite, simple, and undirected. For the adjacency eigenvalues $\lambda_1, \dots, \lambda_n$ of G , put

$$s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2, \quad s^-(G) = \sum_{\lambda_i < 0} \lambda_i^2, \quad \beta(G) = |E(G)| - |V(G)| + 1.$$

A connected graph with $|E(G)| = |V(G)| + 5$ has cyclomatic rank six.

Theorem 1.1 (Conditional main implication—draft only). *Assume both of the following premises.*

1. *The owner-exact multiblock theorem covers every finite simple connected rank-six graph having at least two positive-rank cyclic blocks.*
2. *For every one of the 1198 suppressed rank-six kernels, every physical parity orbit and every required canonical-plus-coordinate target has an exact, disjoint proof owner establishing the budget described in Proposition 6.1; the order-nine and order-ten owners and the orders-2–10 implication master pass full exact replay in both normal and optimized Python modes; and an all-connected synthesis master checks the two scopes as disjoint and exhaustive.*

Then every finite simple connected graph G with $|V(G)| = n$ and $|E(G)| = n + 5$ satisfies

$$s^+(G) \geq n.$$

Remark 1.2 (Current logical status). Theorem 1.1 is an implication, not an unconditional theorem. At the time of this draft the required order-nine/order-ten promotion owners and master acceptance records are not all present. A coverage count, search-completion field, process exit status, or digest without exact replay and exact owner-set equality does not discharge the premise.

No strict inequality, equality classification, edge-addition monotonicity, one-edge-subdivision monotonicity, result for nonsimple realizations, human review, independent verification, peer review, publication, or external dissemination is claimed.

2 Analytic foundations

For a finite graph $H = (V, E)$ define

$$\kappa(H) = \min_{\substack{R \succeq 0 \\ R_{vv} = 1 \ (v \in V)}} \sum_{uv \in E} \frac{2}{1 - R_{uv}}, \quad (1)$$

where a summand is $+\infty$ if $R_{uv} = 1$.

Lemma 2.1 (DNN witness implication). *For every finite simple graph H ,*

$$s^-(H) \leq \kappa(H).$$

Consequently, any feasible correlation matrix with objective at most b proves $s^+(H) \geq 2|E(H)| - b$.

Proof. Write the adjacency matrix as $A = P - N$, where $P, N \succeq 0$ are its positive and negative spectral parts, and put $M = N \circ N$. The Schur product theorem gives $M \succeq 0$ and $M \geq 0$ entrywise. Moreover,

$$\langle J, M \rangle = \text{tr } N^2 = s^-(H), \quad \frac{s^-(H)}{2} = - \sum_{uv \in E(H)} N_{uv} \leq \sum_{uv \in E(H)} \sqrt{M_{uv}}.$$

For every feasible R ,

$$\langle J, M \rangle - 2 \sum_{uv \in E(H)} (1 - R_{uv}) M_{uv} = \langle R, M \rangle + 2 \sum_{\substack{u < v \\ uv \notin E(H)}} (1 - R_{uv}) M_{uv} \geq 0.$$

Weighted Cauchy–Schwarz and cancellation when $s^-(H) > 0$ give

$$s^-(H) \leq \sum_{uv \in E(H)} \frac{2}{1 - R_{uv}}.$$

The zero and infinite-objective cases are immediate. Minimize over R and use $s^+(H) + s^-(H) = \text{tr } A^2 = 2|E(H)|$. $\square$

Lemma 2.2 (Exact path cost). *If a kernel edge with endpoint correlation r is replaced by a path of length $q \geq 1$, its least DNN contribution in excess of its q edges is*

$$f_q(r) = q \tan^2 \left(\frac{\arccos((-1)^q r)}{2q} \right).$$

For fixed r and parity, $f_{q+2}(r) \leq f_q(r)$, with endpoint values understood in the extended sense.

Proof. Represent the endpoints by unit vectors and alternately negate vectors along the path. The transformed endpoint angle is $\gamma = \arccos((-1)^q r)$. The spherical triangle inequality and convexity of $\sec^2(x/2)$ make q equal angles optimal, giving $q \sec^2(\gamma/(2q)) = q + f_q(r)$. With $z = \gamma/(2q)$,

$$\frac{d}{dq} \left[q \tan^2 \left(\frac{\gamma}{2q} \right) \right] = \tan z \sec^2 z (\sin z \cos z - 2z) \leq 0.$$

One-sided limits handle endpoint cases. □

Lemma 2.3 (One-vertex additivity). *If $H = H_1 \vee H_2$ is a one-vertex sum, then $\kappa(H) = \kappa(H_1) + \kappa(H_2)$. In particular, $\kappa(T) = |E(T)|$ for every finite tree T .*

Proof. Restriction gives one inequality. For the other, identify the common unit vector in minimizing Gram representations and place the two perpendicular components in mutually orthogonal subspaces. The objectives add. Every tree edge costs at least one, and opposite unit vectors on the two bipartition classes attain one on each edge. □

3 Block decomposition

Lemma 3.1 (Cyclic-block rank additivity). *The cyclomatic rank of a connected graph is the sum of the cyclomatic ranks of its positive-rank cyclic blocks.*

Proof. Following the block-cut tree, adjoining a block B at one cut vertex adds $|V(B)| - 1$ vertices and $|E(B)|$ edges, and hence adds $|E(B)| - |V(B)| + 1 = \beta(B)$ to the rank. Bridge blocks contribute zero. □

Thus the positive cyclic-block ranks of a rank-six graph form exactly one of

$$\begin{aligned} 1^6, \quad 2 + 1^4, \quad 2 + 2 + 1 + 1, \quad 2 + 2 + 2, \quad 3 + 1 + 1 + 1, \quad 3 + 2 + 1, \quad 3 + 3, \\ 4 + 1 + 1, \quad 4 + 2, \quad 5 + 1, \quad 6. \end{aligned} \tag{2}$$

The first ten rows are the multiblock branch; the final row has one positive-rank cyclic block. This is a structural dichotomy only and does not by itself prove either branch.

4 Verified multiblock interface

Proposition 4.1 (Multiblock branch). *Every finite simple connected rank-six graph with at least two positive-rank cyclic blocks satisfies $s^+(G) \geq |V(G)|$.*

The proof is the owner-exact ledger in the multiblock audit [2]. Its scope consists of the first ten rows of (3). It separates direct DNN rows, structural pre-sieve rows, and the packet residuals A–I. Shared cuts, repeated and nested owners, positive bridge connectors, and arbitrary rooted branches are assigned before credit is charged. The direct rank-five-equality-plus-triangle boundary is nonstrict and is not promoted to a strict claim. This skeleton cites that verified exposition rather than duplicating its long owner-by-owner proof.

5 Unique positive-rank block

Lemma 5.1 (Rank-six suppression). *Let B be a finite simple 2-connected graph with $\beta(B) = 6$. Suppressing every degree-two vertex produces a loopless 2-connected multigraph K of minimum degree at least three such that*

$$|E(K)| = |V(K)| + 5, \quad 2 \leq |V(K)| \leq 10.$$

Conversely, B is a simple realization obtained by replacing kernel edges by positive-length, internally disjoint paths meeting only at prescribed branch endpoints.

Proof. Suppression preserves $|E| - |V|$, connectedness, and rank. A loop would come from a cycle meeting the remainder only at one branch vertex, contradicting 2-connectivity. If $k = |V(K)|$, minimum degree three and $|E(K)| = k + 5$ give $3k \leq 2k + 10$, so $k \leq 10$; loopless 2-connectivity gives $k \geq 2$. Reversing suppression gives the stated path model. Simplicity is a separate required condition, especially in parallel classes. $\square$

Proposition 5.2 (Exact kernel census). *The rank-six suppressed kernels have the following verified distribution.*

$ V(K) $	2	3	4	5	6	7	8	9	10
$\#K$	1	4	26	84	216	314	325	162	66

There are 1198 isomorphism classes in total.

The exact integer census regenerates multiplicity vectors, rejects supports with a cut vertex, canonicalizes under isomorphism, and compares the complete sorted output with `research/fixtures/rank-six-kernels.json`. No floating-point or spectral calculation enters this census. Its reproducible command is listed in Appendix A.

6 Conditional finite interface

For each parallel class of multiplicity m with o odd replacement paths, the canonical simple lengths are

$$c(m, o) = \begin{cases} (2, \dots, 2), & o = 0, \\ (1, \underbrace{3, \dots, 3}_{o-1}, \underbrace{2, \dots, 2}_{m-o}), & o > 0. \end{cases}$$

Simplicity permits at most one unit path in a parallel class. After permuting parallel physical edges, every realization vector ℓ in the parity row satisfies $c_i \leq \ell_i$ and $\ell_i - c_i \in 2\mathbb{Z}_{\geq 0}$.

Proposition 6.1 (Conditional exact-owner interface). *Assume every physical parity orbit of every kernel in Proposition 5.2 is owned either by an exact coarse certificate that remains within excess five under every same-parity lengthening, or by exact certificates for every target*

$$\mathcal{F}(c) = \{c\} \cup \{c + 2\mathbf{e}_i : 1 \leq i \leq |E(K)|\}.$$

Assume that regenerated target keys equal the disjoint union of checked owner keys. Then every simple subdivision of every rank-six kernel, with arbitrary finite rooted trees attached at branch or internal subdivision vertices, satisfies $s^+(G) \geq |V(G)|$.

Proof. If $\ell = c$, use the canonical target. Otherwise choose i with $\ell_i \geq c_i + 2$ and start from the owner of $c + 2\mathbf{e}_i$. Lemma 2.2 permits every remaining same-parity coordinate increase without increasing excess. Hence the subdivided core B , with L edges, has $\kappa(B) \leq L + 5$.

If attached rooted trees contain t edges, Lemma 2.3 gives $\kappa(G) \leq L + 5 + t$. Rank six gives $|V(G)| = L - 5 + t$, and Lemma 2.1 yields

$$s^+(G) \geq 2(L + t) - (L + 5 + t) = L - 5 + t = |V(G)|.$$

Different frontier targets may use different Gram matrices. No parity-changing subdivision or spectral monotonicity is used. $\square$

The premise of Proposition 6.1 is the open promotion gate in this draft. Orders two through eight having theorem material does not authorize silently extending the statement through orders nine and ten. In particular, current coverage verifiers are not substitutes for the missing promotion owners or the fail-closed orders-2–10 master.

7 Conditional assembly

Proof of Theorem 1.1. By Lemma 3.1, the positive block ranks have exactly the eleven forms in (3). If there are at least two positive-rank cyclic blocks, apply Proposition 4.1. Otherwise there is one rank-six cyclic block. Lemma 5.1, Proposition 5.2, and the assumed exact-owner premise allow Proposition 6.1. The two cases are disjoint and exhaustive. This proves the implication, while making no assertion that its currently blocked finite premise has been discharged. $\square$

A Reproducibility ledger and placeholders

A.1 Commands backed by present verified exposition

From the repository root, the exact kernel census and multiblock ledger are intended to be checked in normal and optimized modes:

```
python3 research/rank-six-kernel-census-verifier.py
python3 -O research/rank-six-kernel-census-verifier.py
python3 research/hexacyclic-multiblock-ledger-verifier.py
python3 -O research/hexacyclic-multiblock-ledger-verifier.py
```

The lower single-block owner currently exposed by the repository may likewise be audited with

```
python3 research/rank-six-order2-7-master-verifier.py
python3 -O research/rank-six-order2-7-master-verifier.py
python3 research/rank-six-order-eight-kernel-theorem-verifier.py --full
python3 -O research/rank-six-order-eight-kernel-theorem-verifier.py --full
```

A.2 Blocked promotion checks

The following command is deliberately fail-closed until all required promotion owners and frozen outputs exist:

```
python3 research/rank-six-order2-10-master-verifier.py
python3 -O research/rank-six-order2-10-master-verifier.py
```

Current nonzero exit is a blocker, not a successful theorem test. The order-nine and order-ten coverage commands may report partial exact progress, but their output must not be cited as a theorem owner:

```
python3 research/rank-six-order-nine-coverage-verifier.py
python3 -O research/rank-six-order-nine-coverage-verifier.py
python3 research/rank-six-order-ten-coverage-verifier.py
python3 -O research/rank-six-order-ten-coverage-verifier.py
```

Promotion placeholders (intentionally unresolved). Before the conditional label can be removed, replace each item below with a committed, independently regenerated record checked against the manuscript.

1. **TODO-ORDER9-OWNER:** theorem-owner path, schema, source SHA-256, canonical full-replay output SHA-256, exact owner counts, and normal/optimized byte-identity record.
2. **TODO-ORDER10-OWNER:** theorem-owner path, schema, source SHA-256, canonical full-replay output SHA-256, exact owner counts, and normal/optimized byte-identity record.
3. **TODO-RANK6-MASTER:** green orders-2–10 implication manifest, all four child-owner identities, census identity, transitive root, and hostile mutation report.
4. **TODO-ALL-CONNECTED-MASTER:** green two-branch synthesis manifest, scope-disjointness audit, exact conclusion, transitive root, and hostile mutation report.
5. **TODO-CLEAN-REPLAY:** interpreter version, platform, commands, wall-clock/storage notes, complete logs, and normal versus optimized output digests from a clean artifact directory.

No placeholder is evidence, and no digest by itself proves a mathematical predicate.

A.3 Build

This draft is compiled from the repository root with

```
bash scripts/build-paper.sh all-hexacyclic-graphs
```

Build success checks only the LaTeX artifact. It does not turn a red finite gate green and does not promote Theorem 1.1.

B Certificate schemas to freeze at promotion

The eventual machine-readable appendix must specify, without relying on prose status fields:

1. canonical ASCII encodings and exact kernel/parity/frontier key schemas;
2. rational Gram-chain records, positive denominator checks, exact unit and shared-endpoint identities, and the integer predicate “excess at most five”;
3. symbolic records with exact positive-semidefinite and path-cost checks;
4. structural records with typed induced territories, unique ownership, and no relabelling as DNN witnesses;

5. complete regenerated-key equality, duplicate/extra/missing-key rejection, and disjoint final ownership;
6. source, fixture, chunk, canonical-output, and transitive manifest hashes, with a statement that hashes identify bytes rather than prove predicates.

This section is a schema checklist only. It asserts no present order-nine or order-ten certificate completion.

C Trusted base and limitations

The analytic trust base consists of the displayed matrix arguments, exact path elimination, block decomposition, and finite-kernel reduction. The computer-assisted trust base additionally includes the committed verifier sources, exact integer and rational semantics, canonical parsing and labelling, the Python runtime, and executing hardware. Numerical optimization may be used to discover witnesses but is not a proof object. Rooted-tree attachment means a genuine one-vertex sum; a connector meeting a core twice is outside that lemma. Kernel parallel edges are permitted only before realization; the graph to which the trace identity is applied remains simple.

AI disclosure and nonpublication notice

This draft was generated by an AI coding and mathematical reasoning system from the repository materials cited below. No human authorship, human review, independent verification, peer review, acceptance, publication, or theorem announcement is claimed. The manuscript is a conditional internal skeleton pending the explicit promotion gates in Appendix A.

References

- [1] *LaTeX exposition modules for the rank-six synthesis*, repository proof note, 2026; `positive-square-energy/hexacyclic-general/latex-exposition-modules.md`.
- [2] *Hexacyclic multiblock theorem: exhaustive ledger and hostile audit*, repository proof note, 2026; `positive-square-energy/hexacyclic-general/multiblock-items1-7-combined-theorem-audit.md`.
- [3] *Audit of the rank-six single-block implication*, repository proof note, 2026; `positive-square-energy/hexacyclic-general/rank-six-single-block-implication-audit.md`.
- [4] *Rank-six orders 2–10 master architecture*, promotion-gated repository note, 2026; `positive-square-energy/hexacyclic-general/rank-six-order2-10-master-architecture.md`.
- [5] *Order-ten rank-six master architecture*, completion-gated repository note, 2026; `positive-square-energy/hexacyclic-general/order-ten-rank-six-master-architecture.md`.